



Qualification Pack



Digital Construction (WorldSkills)

QP Code: CON/Q4106

Version: 1.0

NSQF Level: 4.5

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CON/Q4106: Digital Construction (WorldSkills)

Brief Job Description

Digital Construction using Building Information Modelling (BIM) is a process for creating and managing information on a construction project across the project lifecycle. One of the key outputs of this process is the digital Building Information Model, the digital description of every aspect of the built asset. This digital model draws on information assembled collaboratively and updated at key stages of a project. Creating a digital Building Information Model enables those who interact with the building to optimise their actions, resulting in a more excellent whole life value for the asset.

Personal Attributes

The person should be physically fit to work in varying locations and conditions. The person should be organized, diligent, methodical, safety-conscious, and a prompt decision-maker. In addition to being a team player, the individual should have good communication skills

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [CON/N4101: Work Organization, Management and Communication](#)
2. [CON/N4111: Preparation for the Test Project](#)
3. [CON/N4112: Drawing of Test Project \(Digital Construction\)](#)
4. [CON/N4105: Quality Check and Handover](#)

Qualification Pack (QP) Parameters

Sector	Construction
Sub-Sector	Real Estate and Infrastructure construction
Occupation	For Skill Competitions
Country	India
NSQF Level	4.5
Credits	17
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7122.0100



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Minimum Educational Qualification & Experience	Basic Literacy and Numeracy (The Competitors must not be older than 25 years in the year of the Competition.)
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	14 Years
Last Reviewed On	NA
Next Review Date	31/01/2026
NSQC Approval Date	31/01/2024
Version	1.0
Reference code on NQR	QG-4.5-CO-01794-2024-V1-CSDCI
NQR Version	1.0

Remarks:

i) There is no specified minimum job entry age for the skill competitions, however, the competitor(s) must not be older than 22 years in the year of the competition as per WorldSkills Standards. ii) The validity of these qualifications shall be only for two years or as per the WorldSkills competition cycle, whichever is earlier". (Please refer clause 4.1.(d) of NCVET Order No. 32001/06/2023/NCVET, dated 26.01.2024)



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CON/N4101: Work Organization, Management and Communication

Description

This NOS deals with ensuring safe and healthy working environment at competition area by proper utilization of safety gears and effective communication skill with juries and competitors.

Scope

The scope covers the following :

- Adopt healthy and safe work practices at competition area
- Management of workplace
- Effective communication at competition area

Elements and Performance Criteria

Adopt healthy and safe work practices at competition area

To be competent, the user/individual on the job must be able to:

- PC1.** Explain health and safety legislation, obligations, and documentation
- PC2.** check and ensure the availability of appropriate and usable Personal Protective Equipment (PPE) at the workplace
- PC3.** use the appropriate PPE according to the nature of work and ensure other personnel also do the same
- PC4.** follow and promote the appropriate measures to ensure hygiene in the workplace, e.g.regular cleaning and sanitization
- PC5.** follow the applicable emergency procedures and ensure the personnel also do the same
- PC6.** Learn precautions for the safe use of power and cordless tools
- PC7.** Apply standards and laws relating to security, safety, and hygiene in plastering and drywall systems

Management of workplace

To be competent, the user/individual on the job must be able to:

- PC8.** establish methods for optimal and economically organized construction site. regarding construction plan and equipment, organization and procedures, material management, staffing and timelines
- PC9.** determine the correct calculations and estimates by using principles and methods
- PC10.** store plasterboards and related products safely and securely
- PC11.** be proactive in own continuous professional development in order to keep abreast of methods of working in the construction industry and changing technologies, for example acoustics, sustainability and environmental impact
- PC12.** Work effectively as part of teams
- PC13.** Work effectively with other trades
- PC14.** Establish and maintain high quality standards and working processes
- PC15.** identify problems promptly and manage their resolution



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- PC16.** select, use and store all materials safely
- PC17.** plan work areas to maximize efficiency and maintain the discipline of regular tidying
- PC18.** principles and methods for determine the correct calculations and estimates
- PC19.** understand technical terminology and symbols

Effective communication at competition area

To be competent, the user/individual on the job must be able to:

- PC20.** visualize and translate customer wishes making recommendations which meet/improve their design and budgetary requirements where qualified to do so
- PC21.** Produce a cost and time estimate for customers
- PC22.** introduce related trades to support customer requirements
- PC23.** work effectively in a team to facilitate efficiency/productivity/quality and cost control
- PC24.** recognize the needs of architects and related trades
- PC25.** explain the importance of swiftly resolving misunderstandings and conflicting demands

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Health, hygiene, and safety legislation, obligations, regulations and documentation
- KU2.** principles of working safely with electricity
- KU3.** Accident/first-aid/fire/emergency procedures and reporting
- KU4.** situations when personal protective equipment must be used
- KU5.** purposes, uses, care, maintenance, and storage of all hand and powered tools and equipment together with their safety implications
- KU6.** purposes, uses, care and storage of materials
- KU7.** sustainability measures applying to the use of green materials and recycling
- KU8.** the significance of planning, accuracy, checking, and attention to detail in all working practices
- KU9.** importance of integrity and trustworthiness
- KU10.** value of managing own continuing professional development
- KU11.** value of building and maintaining trust and productive working relationships
- KU12.** importance of swiftly resolving misunderstandings and conflicting demands
- KU13.** need for security for the storage of tools and materials
- KU14.** criteria for being understandable within teams and to non specialists
- KU15.** recognised IT systems and related professional design software

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain the appropriate data and records
- GS2.** read the appropriate reports and literature concerning the test project



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- GS3.** communicate professionally with all the stakeholders
- GS4.** listen attentively to understand the information/ instructions being shared and take appropriate action
- GS5.** coordinate with co-workers to achieve the work objectives
- GS6.** plan and execute tasks based on priority
- GS7.** identify possible disruptions to work and take appropriate mitigation measures
- GS8.** take prompt action to deal with workplace emergencies and accidents
- GS9.** evaluate all possible solutions to work-related problems and select the best one
- GS10.** follow the recommended practices for the timely completion of work and achievement of organizational objectives



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Adopt healthy and safe work practices at competition area</i>	10	25	-	-
PC1. Explain health and safety legislation, obligations, and documentation	-	-	-	-
PC2. check and ensure the availability of appropriate and usable Personal Protective Equipment (PPE) at the workplace	-	-	-	-
PC3. use the appropriate PPE according to the nature of work and ensure other personnel also do the same	-	-	-	-
PC4. follow and promote the appropriate measures to ensure hygiene in the workplace, e.g.regular cleaning and sanitization	-	-	-	-
PC5. follow the applicable emergency procedures and ensure the personnel also do the same	-	-	-	-
PC6. Learn precautions for the safe use of power and cordless tools	-	-	-	-
PC7. Apply standards and laws relating to security, safety, and hygiene in plastering and drywall systems	-	-	-	-
<i>Management of workplace</i>	10	25	-	-
PC8. establish methods for optimal and economically organized construction site. regarding construction plan and equipment, organization and procedures, material management, staffing and timelines	-	-	-	-
PC9. determine the correct calculations and estimates by using principles and methods	-	-	-	-
PC10. store plasterboards and related products safely and securely	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. be proactive in own continuous professional development in order to keep abreast of methods of working in the construction industry and changing technologies, for example acoustics, sustainability and environmental impact	-	-	-	-
PC12. Work effectively as part of teams	-	-	-	-
PC13. Work effectively with other trades	-	-	-	-
PC14. Establish and maintain high quality standards and working processes	-	-	-	-
PC15. identify problems promptly and manage their resolution	-	-	-	-
PC16. select, use and store all materials safely	-	-	-	-
PC17. plan work areas to maximize efficiency and maintain the discipline of regular tidying	-	-	-	-
PC18. principles and methods for determine the correct calculations and estimates	-	-	-	-
PC19. understand technical terminology and symbols	-	-	-	-
<i>Effective communication at competition area</i>	10	20	-	-
PC20. visualize and translate customer wishes making recommendations which meet/improve their design and budgetary requirements where qualified to do so	-	-	-	-
PC21. Produce a cost and time estimate for customers	-	-	-	-
PC22. introduce related trades to support customer requirements	-	-	-	-
PC23. work effectively in a team to facilitate efficiency/productivity/quality and cost control	-	-	-	-
PC24. recognize the needs of architects and related trades	-	-	-	-
PC25. explain the importance of swiftly resolving misunderstandings and conflicting demands	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	30	70	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	CON/N4101
NOS Name	Work Organization, Management and Communication
Sector	Construction
Sub-Sector	Real Estate and Infrastructure construction
Occupation	For Skill Competitions
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2026
NSQC Clearance Date	31/01/2024



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CON/N4111: Preparation for the Test Project

Description

This NOS deals with the preparation of test project for Digital Construction Skill.

Scope

The scope covers the following :

- Understand the hardware and software
- Carryout interpretation of the client brief

Elements and Performance Criteria

Understand the hardware and software

To be competent, the user/individual on the job must be able to:

- PC1.** Power up the equipment and activate the appropriate modelling software
- PC2.** Set up and check peripheral devices such as keyboard, and mouse
- PC3.** Use computer operating systems and specialist software to create and manage and store files proficiently both locally and to the Common Data environment BIM project
- PC4.** Select correct drawing packages from an on-screen menu or graphical equivalent
- PC5.** Use various techniques for accessing and using BIM software such as a mouse, menu, or tool bar
- PC6.** Configure the parameters of the software

Carryout interpretation of the client brief

To be competent, the user/individual on the job must be able to:

- PC7.** Interpret clients' briefs to determine:
- PC8.** Outline requirements of each project
- PC9.** Client goals
- PC10.** Locations
- PC11.** Work from BEPs, client briefs and EIRs to address client and project requirements
- PC12.** Create and edit BIM information within CDEs as per BEPs across the lifecycle of construction projects and provide access/set permissions to the necessary folders to the BIM team

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Computer operating systems to be able to use and manage computer files and software correctly
- KU2.** Peripheral devices used in the digital construction process
- KU3.** Specific specialist technical operations within design software
- KU4.** The workflow for digital construction projects



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- KU5.** The limitations of the design software
- KU6.** Formats and resolutions
- KU7.** The role of energy-efficient hardware and software in digital construction and their contribution towards UN SDG 7 (Affordable and Clean Energy)
- KU8.** What information is provided in a client's brief
- KU9.** The importance of the Exchange Information Requirements (EIR)
- KU10.** The importance of the Asset Information Requirements (AIR) of the project
- KU11.** The relevant and current industry standards
- KU12.** How to work from a BIM Execution Plan (BEP) from the EIR
- KU13.** How to create and edit BIM information within a Common Data Environment (CDE) across the lifecycle of construction
- KU14.** The importance of file structures and sharing protocols within the CDE
- KU15.** How client briefs may incorporate elements related to UN SDGs such as SDG 3 (Good Health and Well-being), SDG 7 (Affordable and Clean Energy), SDG 8 (Decent Work and Economic Growth), SDG 11 (Sustainable Cities and Communities), and SDG 12 (Responsible Consumption and Production)

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** write in at least one language, preferably in the local language of the site
- GS2.** read sketches/routine work drawings, instructions provided for the work and various signboards, safety rules, safety tags, exit route information in one or more languages, preferably in the local language of the site
- GS3.** speak in one or more language, preferably one of the local language at the site
- GS4.** communicate orally and effectively with team members
- GS5.** analyze the safety aspects of the workplace
- GS6.** plan work and organize required resource effectively
- GS7.** complete work as per agreed time schedule and quality parameters
- GS8.** resolve any conflict within the teammates
- GS9.** evaluate the complexity of the tasks
- GS10.** identify any violation of safety norms during the work



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Understand the hardware and software</i>	15	35	-	-
PC1. Power up the equipment and activate the appropriate modelling software	-	-	-	-
PC2. Set up and check peripheral devices such as keyboard, and mouse	-	-	-	-
PC3. Use computer operating systems and specialist software to create and manage and store files proficiently both locally and to the Common Data environment BIM project	-	-	-	-
PC4. Select correct drawing packages from an on-screen menu or graphical equivalent	-	-	-	-
PC5. Use various techniques for accessing and using BIM software such as a mouse, menu, or tool bar	-	-	-	-
PC6. Configure the parameters of the software	-	-	-	-
<i>Carryout interpretation of the client brief</i>	15	35	-	-
PC7. Interpret clients' briefs to determine:	-	-	-	-
PC8. Outline requirements of each project	-	-	-	-
PC9. Client goals	-	-	-	-
PC10. Locations	-	-	-	-
PC11. Work from BEPs, client briefs and EIRs to address client and project requirements	-	-	-	-
PC12. Create and edit BIM information within CDEs as per BEPs across the lifecycle of construction projects and provide access/set permissions to the necessary folders to the BIM team	-	-	-	-
NOS Total	30	70	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	CON/N4111
NOS Name	Preparation for the Test Project
Sector	Construction
Sub-Sector	Real Estate and Infrastructure construction
Occupation	For Skill Competitions
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2026
NSQC Clearance Date	31/01/2024



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CON/N4112: Drawing of Test Project (Digital Construction)

Description

This NOS deals with the drawing of test project for Digital Construction Skill.

Scope

The scope covers the following :

- Building Information Modelling
- Model coordination
- Asset information modelling
- Data creation and management
- Site execution

Elements and Performance Criteria

Building Information Modelling

To be competent, the user/individual on the job must be able to:

- PC1.** Open an appropriate Project Information Model from the relevant directory within the CDE
- PC2.** Populate Project Properties with given information
- PC3.** Set each model up as collaborative files
- PC4.** Create work sets
- PC5.** Set each project Location, orientation and level datum
- PC6.** Create each structural grid
- PC7.** Create BIM models as per given drawings
- PC8.** Save each BIM model with a prescribed starting View
- PC9.** Save each Project Information Model within the CDE for use by the other disciplines via Construction Cloud software
- PC10.** Adhere to the requirements of the BEP to ensure Data drops are made via the Construction Cloud software
- PC11.** Produce scaled detailed drawings to the required standard using callout and details items
- PC12.** Create 3D visuals to illustrate each building from different viewpoints
- PC13.** Apply problem-solving techniques to overcome challenges that arise during the BIM modelling process
- PC14.** Use creative and innovative thinking to propose improvements in the BIM process and to solve technical and design problems
- PC15.** Stay abreast of and apply the latest trends and advancements in BIM technology to increase efficiency and quality of models

Model coordination

To be competent, the user/individual on the job must be able to:

- PC16.** Federate Structural, Architectural and specialist designer project models
- PC17.** Perform Clash Inspections as per the BEP



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- PC18.** Export all tests as per the BEP and issue to the CDE
- PC19.** Save and issue federated files as per the BEP
- PC20.** Quality assure each federated project model by visually inspecting each CDE hosted model
- PC21.** Identify issues with coordination in each new build that haven't shown up in the three clash tests
- PC22.** For each issue discovered: Create the issue, Add annotation describing the issue, Assign the issue to the BIM Manager on the project, Name each view as per the BEP

Asset information modelling

To be competent, the user/individual on the job must be able to:

- PC23.** Update Project Information Models from the published directory
- PC24.** Ensure all required assets have the required data populated as per the latest standard
- PC25.** Add classification information to each model elements – referring to the project BEP
- PC26.** Create COBie and IFC data from BIM model elements

Data creation and management

To be competent, the user/individual on the job must be able to:

- PC27.** Create Shared Parameter files with custom parameters for selected building elements
- PC28.** Create Custom Tags to visually express technical information from custom parameters
- PC29.** Create colour filters to visually express technical information from the custom parameters on duplicate plans, sections and 3D cut sections
- PC30.** Create schedules of project information including custom parameters
- PC31.** Plot PDF sheet sets of combined PDFs to correct scale and correct sheet sizes
- PC32.** Run visual scripts to automate data extraction
- PC33.** Visualise data with dashboards within the CDE
- PC34.** Take off quantities from PDF and 3D models
- PC35.** Incorporate sustainability considerations into the data creation and management process
- PC36.** Use BIM to support decision-making that reduces the environmental impact of construction projects
- PC37.** Carry out lifecycle analysis as part of the BIM process to support sustainable decision-making

Site execution

To be competent, the user/individual on the job must be able to:

- PC38.** Publish construction documents for construction
- PC39.** Create and complete digital safety checklists for construction site
- PC40.** Track construction site issues and assign them to stakeholders
- PC41.** Set up meetings, protocol outcomes, and assign action items to meeting attendees
- PC42.** Manage assets and track assets for construction sites

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Programmes used in the BIM authoring and collaboration process
- KU2.** Computer operating systems in order to use and manage computer files and software



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- KU3.** The importance of organising BIM objects into meaningful groups of disciplinary information that can be managed visually
- KU4.** How to create Building Information Models (BIMs), (Structural and Architectural)
- KU5.** Principles of Technical Design
- KU6.** How to access and use documentation in a BIM project
- KU7.** How to set up a BIM as a collaborative file
- KU8.** How to set up a project location, orientation, and level datum
- KU9.** The use of Work in Progress (WIP) folders or information states
- KU10.** The importance of Information exchanges (Data drops) at key project stages and of working to the requirements of the BEP
- KU11.** How to produce a given detail to current standards
- KU12.** The use of 3D visualisation tools
- KU13.** The principles of creative thinking and problem-solving techniques
- KU14.** The potential benefits and risks associated with innovative approaches
- KU15.** The latest trends and advancements in the field of BIM technology
- KU16.** How to create Building Information Models (BIMs) that are aligned with SDGs, particularly SDGs 3, 7, 8, 11, and 12. For instance, models could include features promoting health and well-being, energy efficiency, sustainable urban planning, and responsible consumption and production
- KU17.** How to federate different discipline models with the same model format
- KU18.** What a hard and soft clash is and how to use the BEP to ensure requirements/responsibilities are achieved and perform a clash inspection
- KU19.** How to upload and report issues to BIM project and the CDE
- KU20.** How to perform and record details of a visual clash inspection
- KU21.** How the coordination of different models can contribute to the achievement of the SDGs by reducing waste and inefficiencies (SDG 12), promoting energy efficiency (SDG 7), or creating healthier environments (SDG 3)
- KU22.** How to use Current Documentation standards for Building Information Modelling
- KU23.** How to populate each model with structural asset data
- KU24.** How to apply Classification Standards for model elements
- KU25.** How to create IFC data sheets
- KU26.** The importance of data creation and extraction from the digital model for use by stakeholders in the project
- KU27.** How to create a Shared Parameter file for custom data requirements
- KU28.** How to create schedules of project information with customised data fields
- KU29.** How to use filters with parameters to visually express custom data requirements
- KU30.** How to create a visualisation that express's statutory regulations around fire and/or thermal u-values or similar
- KU31.** How to plot a sheet to PDF format
- KU32.** Basic understanding and ability in visual programming
- KU33.** The use of ACC Insight to visualise data within the CDE
- KU34.** How to use quantity take off tools



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- KU35.** The concept of sustainability in digital construction and the role of BIM in supporting sustainable practices
- KU36.** The potential impact of construction projects on the environment and how BIM can help to mitigate these impacts
- KU37.** The principles of lifecycle analysis and how this can be incorporated into the BIM process
- KU38.** How data extraction from digital models can be used to track and improve performance against the SDGs
- KU39.** How to publish construction documents for construction
- KU40.** How to create and complete digital safety checklists for construction sites
- KU41.** How to track construction site issues and assign them to stakeholders
- KU42.** How to set up meetings, protocol and outcomes, and assign action items to meeting attendee
- KU43.** How to manage assets and track assets for construction sites

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** write in at least one language, preferably in the local language of the site
- GS2.** read sketches/routine work drawings, instructions provided for the work and various signboards, safety rules, safety tags, exit route information in one or more languages, preferably in the local language of the site
- GS3.** speak in one or more language, preferably one of the local language at the site
- GS4.** communicate orally and effectively with team members
- GS5.** analyze the safety aspects of the workplace
- GS6.** plan work and organize required resource effectively
- GS7.** complete work as per agreed time schedule and quality parameters
- GS8.** resolve any conflict within the teammates
- GS9.** evaluate the complexity of the tasks
- GS10.** identify any violation of safety norms during the work



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Building Information Modelling</i>	6	14	-	-
PC1. Open an appropriate Project Information Model from the relevant directory within the CDE	-	-	-	-
PC2. Populate Project Properties with given information	-	-	-	-
PC3. Set each model up as collaborative files	-	-	-	-
PC4. Create work sets	-	-	-	-
PC5. Set each project Location, orientation and level datum	-	-	-	-
PC6. Create each structural grid	-	-	-	-
PC7. Create BIM models as per given drawings	-	-	-	-
PC8. Save each BIM model with a prescribed starting View	-	-	-	-
PC9. Save each Project Information Model within the CDE for use by the other disciplines via Construction Cloud software	-	-	-	-
PC10. Adhere to the requirements of the BEP to ensure Data drops are made via the Construction Cloud software	-	-	-	-
PC11. Produce scaled detailed drawings to the required standard using callout and details items	-	-	-	-
PC12. Create 3D visuals to illustrate each building from different viewpoints	-	-	-	-
PC13. Apply problem-solving techniques to overcome challenges that arise during the BIM modelling process	-	-	-	-
PC14. Use creative and innovative thinking to propose improvements in the BIM process and to solve technical and design problems	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. Stay abreast of and apply the latest trends and advancements in BIM technology to increase efficiency and quality of models	-	-	-	-
<i>Model coordination</i>	6	14	-	-
PC16. Federate Structural, Architectural and specialist designer project models	-	-	-	-
PC17. Perform Clash Inspections as per the BEP	-	-	-	-
PC18. Export all tests as per the BEP and issue to the CDE	-	-	-	-
PC19. Save and issue federated files as per the BEP	-	-	-	-
PC20. Quality assure each federated project model by visually inspecting each CDE hosted model	-	-	-	-
PC21. Identify issues with coordination in each new build that haven't shown up in the three clash tests	-	-	-	-
PC22. For each issue discovered: Create the issue, Add annotation describing the issue, Assign the issue to the BIM Manager on the project, Name each view as per the BEP	-	-	-	-
<i>Asset information modelling</i>	6	14	-	-
PC23. Update Project Information Models from the published directory	-	-	-	-
PC24. Ensure all required assets have the required data populated as per the latest standard	-	-	-	-
PC25. Add classification information to each model elements - referring to the project BEP	-	-	-	-
PC26. Create COBie and IFC data from BIM model elements	-	-	-	-
<i>Data creation and management</i>	6	14	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC27. Create Shared Parameter files with custom parameters for selected building elements	-	-	-	-
PC28. Create Custom Tags to visually express technical information from custom parameters	-	-	-	-
PC29. Create colour filters to visually express technical information from the custom parameters on duplicate plans, sections and 3D cut sections	-	-	-	-
PC30. Create schedules of project information including custom parameters	-	-	-	-
PC31. Plot PDF sheet sets of combined PDFs to correct scale and correct sheet sizes	-	-	-	-
PC32. Run visual scripts to automate data extraction	-	-	-	-
PC33. Visualise data with dashboards within the CDE	-	-	-	-
PC34. Take off quantities from PDF and 3D models	-	-	-	-
PC35. Incorporate sustainability considerations into the data creation and management process	-	-	-	-
PC36. Use BIM to support decision-making that reduces the environmental impact of construction projects	-	-	-	-
PC37. Carry out lifecycle analysis as part of the BIM process to support sustainable decision-making	-	-	-	-
<i>Site execution</i>	6	14	-	-
PC38. Publish construction documents for construction	-	-	-	-
PC39. Create and complete digital safety checklists for construction site	-	-	-	-
PC40. Track construction site issues and assign them to stakeholders	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC41. Set up meetings, protocol outcomes, and assign action Items to meeting attendees	-	-	-	-
PC42. Manage assets and track assets for construction sites	-	-	-	-
NOS Total	30	70	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	CON/N4112
NOS Name	Drawing of Test Project (Digital Construction)
Sector	Construction
Sub-Sector	Real Estate and Infrastructure construction
Occupation	For Skill Competitions
NSQF Level	4.5
Credits	9
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2026
NSQC Clearance Date	31/01/2024



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CON/N4105: Quality Check and Handover

Description

This NOS deals with the quality check of test project prepared as per standards and handing over the same to the concerned juries.

Scope

The scope covers the following :

- Quality check of the test project
- Handover the project to juries

Elements and Performance Criteria

Quality check of the test project

To be competent, the user/individual on the job must be able to:

- PC1.** Inspect equipment, structures, and/or material to identify the nature and causes of errors, defects, or problems
- PC2.** Think critically by using logic and reasoning to identify the strengths and weaknesses of alternative solutions, conclusions or approaches to problems
- PC3.** Identify actual and potential problems
- PC4.** Analyse information and evaluate options to choose and use the best solution
- PC5.** Evaluate solutions and optimize results
- PC6.** The need for all work to be presented to meet customer and related trades needs and expectations
- PC7.** The importance of joint finishing in line with the specifications provided
- PC8.** The different techniques of applying different joint finishes

Handover the project to juries

To be competent, the user/individual on the job must be able to:

- PC9.** understand the process of marking and assessment criteria of respective trade
- PC10.** respond to the juries on the queries being raised by them
- PC11.** Make and follow through decision
- PC12.** Apply appropriate techniques according to buildings' history and fabrication, whilst maintaining their integrity for both internal and external surfaces
- PC13.** Leave work areas in a suitable condition for inspection and subsequent work
- PC14.** Report positive and negative variances in work processes and results, together with their implications
- PC15.** Organize any waste material so that it can be disposed of or recycled efficiently

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:



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- KU1.** The need for all work to be presented to meet customer and related trades needs and expectations
- KU2.** The importance of joint finishing in line with the specifications provided
- KU3.** Presentation including the brushing and cleaning of brickwork plus the tidying and cleaning of the work area (specific skill)
- KU4.** The different techniques of applying different joint finishes
- KU5.** The required quality standards for the task in hand
- KU6.** The nature and causes of substandard work and defects
- KU7.** The available range of quality checks and methods Alternative methods for effecting remedies and repairs

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain work-related notes and records
- GS2.** read the relevant literature to learn about the latest developments in the field of work
- GS3.** listen attentively to understand the information/ instructions being shared by the speaker
- GS4.** communicate clearly and politely with co-workers and clients
- GS5.** coordinate with co-workers to achieve work objectives
- GS6.** plan and prioritize tasks to ensure timely completion
- GS7.** identify possible disruptions to work and take appropriate preventive measures
- GS8.** take quick decisions to deal with workplace emergencies/ accidents
- GS9.** evaluate all possible solutions to a problem to select the best one



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Quality check of the test project</i>	20	48	-	-
PC1. Inspect equipment, structures, and/or material to identify the nature and causes of errors, defects, or problems	-	-	-	-
PC2. Think critically by using logic and reasoning to identify the strengths and weaknesses of alternative solutions, conclusions or approaches to problems	-	-	-	-
PC3. Identify actual and potential problems	-	-	-	-
PC4. Analyse information and evaluate options to choose and use the best solution	-	-	-	-
PC5. Evaluate solutions and optimize results	-	-	-	-
PC6. The need for all work to be presented to meet customer and related trades needs and expectations	-	-	-	-
PC7. The importance of joint finishing in line with the specifications provided	-	-	-	-
PC8. The different techniques of applying different joint finishes	-	-	-	-
<i>Handover the project to juries</i>	10	20	-	-
PC9. understand the process of marking and assessment criteria of respective trade	-	-	-	-
PC10. respond to the juries on the queries being raised by them	-	-	-	-
PC11. Make and follow through decision	-	-	-	-
PC12. Apply appropriate techniques according to buildings' history and fabrication, whilst maintaining their integrity for both internal and external surfaces	-	-	-	-
PC13. Leave work areas in a suitable condition for inspection and subsequent work	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. Report positive and negative variances in work processes and results, together with their implications	-	-	-	-
PC15. Organize any waste material so that it can be disposed of or recycled efficiently	-	-	-	-
NOS Total	30	70	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	CON/N4105
NOS Name	Quality Check and Handover
Sector	Construction
Sub-Sector	Real Estate and Infrastructure construction
Occupation	For Skill Competitions
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2026
NSQC Clearance Date	31/01/2024

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

- The primary aim of the Skill assessment strategy and procedures is to ensure a fair, transparent, objective, and reliable assessment process, promoting excellence in skills and providing a level playing field for all Competitors.
- This skill competition is classed as "fault finding" on all days. Therefore, no Expert and Competitor communication during the competition time, including breaks and lunch period, are allowed. For official Communication, Competition Rules 7.3.3 is strictly applied.
- In accordance with WorldSkills Rules and guidance, on C-3 Mandatory Assessment Training will include practical assessments of each Experts' expertise both technically, and in assessment and marking. Following this, the Chief Expert and Skill Competition Manager will determine who will assess, and who will have the opportunity to enhance their expertise through observation.
- Marking Teams are based on Sub Criteria, which in turn will reflect the weightings in the WorldSkills Occupational Standards (WSOS). The criteria for determining which Marking Team will mark each Sub Criteria is in the Competition's assessment plan as well as the organization and timing of each module.
- To ensure the highest standards of marking consistency, a periodic review of marked tasks will be



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conducted by the Chief Expert or designated senior Experts. Any discrepancies in marking will be discussed and resolved collectively.

- One Marking Team must mark every aspect within the sub criterion.
- Where appropriate, technology-assisted assessment tools may be utilized to ensure consistency and accuracy in the marking process. However, the final decision on marks awarded will always lie with the Marking Teams.
- In accordance with the Rules, there is no blind marking and no compatriot marking. The composition of each Marking Team will ensure that these restrictions are adhered to.
- All Experts involved in the assessment process must declare any potential conflicts of interest. Steps will be taken to ensure that no Expert is placed in a position where their impartiality could be compromised.
- All assessment decisions, especially those requiring deliberation, will be documented meticulously to ensure transparency and to serve as a reference for any post-competition reviews or appeals.
- Exceptions to the rules are permissible only with the agreement of the Chair and Vice Chair of the Competitions Committee. The Skill Advisor must be contacted if this possibility is raised due to a lack of Experts with the required expertise.
- 0: performance below industry standard, 1: performance meets industry standard, 2: performance meets and, in specific respects, exceeds industry standard, 3: performance wholly exceeds industry standard and is judged as excellent

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Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
CON/N4101.Work Organization, Management and Communication	30	70	-	-	100	10



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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
CON/N4111.Preparation for the Test Project	30	70	0	0	100	15
CON/N4112.Drawing of Test Project (Digital Construction)	30	70	0	0	100	60
CON/N4105.Quality Check and Handover	30	70	-	-	100	15
Total	120	280	-	-	400	100



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Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training



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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.



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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.